Palm OS Programming Recipes

Accessing a Handheld Database with the Sync Manager API

This recipe shows you how to access a handheld database by calling the Sync Manager API in your conduit. The Sync Manager API is a low-level interface with which you can implement your own synchronization logic. (Alternatively, you may use one of the conduit frameworks, which implement Palm's standard synchronization logic, provided in the Palm™ Conduit Development Kit.)

This recipe accesses the Date Book database on the handheld using Sync Manager API functions in the following steps:

1. Call `SyncRegisterConduit` to register the conduit for Sync Manager access.
2. Call `SyncOpenDB` to open the handheld database (or optionally create the database).
3. Call Sync Manager API functions to access and edit the handheld database.
4. Call `SyncCloseDB` to allow other conduits access to handheld databases.
5. Call `SyncUnRegisterConduit` to release Sync Manager access to other conduits.

1. Call SyncRegisterConduit

The following code shows declarations used in this and the following steps and then shows how to register your conduit with the Sync Manager:

```
#define DB_NAME "DatebookDB"
#define LOG_STRING "%s returned %d"

CONDHANDLE hSyncHandle;  // handle to use Sync Manager functions
BYTE hDataBase;          // handle to the remote database
WORD rNumRecs;           // number of records in the database

// First, we must register the conduit to use Sync Manager calls.
if (retval = SyncRegisterConduit(hSyncHandle))
{
    // SyncRegisterConduit failed: Handle the error code.
    return (retval);
}
```

2. Call SyncOpenDB

You must open (or create) a database before you can access it.

```
// Open the DatabookDB database.
if (retval = SyncOpenDB(DB_NAME,0,hDataBase,
        (eDbRead|eDbWrite|eDbShowSecret)))
{
    // SyncOpenDB failed: Handle the error code.
    // TODO - If the database does not exist, create it.
    return (retval);
}
```

3. Call Sync Manager API Functions

With the database open, you can now access it. The code below shows a simple example of getting the number of records in the database and printing the result to the HotSync(R) log. At this point, you can implement your synchronization logic.

```
// Retrieve the record count for the DatebookDB.
if (retval = SyncGetDBRecordCount(hDataBase,rNumRecs))
{
    // SyncGetDBRecordCount failed: Handle the error code.
    return (retval);
}

// The call to SyncGetDBRecordCount was successful, write the
// result to log.
if (retval = LogAddFormattedEntry(slText, true, LOG_STRING,
    "LogAddFormattedEntry", rNumRecs))
{
    // LogAddFormattedEntry failed: Handle the error code.
    return (retval);
}
```

4. Call SyncCloseDB

After you finish with a database, you must close it so that another conduit (or handheld application) can access it.

```
// The Sync Manager allows only one database to be open at a time,
// so close the database.
if (retval = SyncCloseDB(hDataBase))
{
    // SyncCloseDB failed: Handle the error code.
    return (retval);
}
```

5. Call SyncUnRegisterConduit

Before your conduit completes, you must unregister it so that other conduits can call the Sync Manager.

```
// Unregister this conduit so that the next conduit
// can use the Sync Manager.
if (retval = SyncUnRegisterConduit(hSyncHandle))
{
    // SyncUnRegisterConduit failed: Handle the error code.
    return (retval);
}
```

Summary

You should now have a basic understanding of the usage of the Sync Manager API for accessing handheld databases. The next phase of the development process should be the definition of a synchronization strategy and a record format for the handheld and the desktop computer.

NOTE: For the sake of simplicity and focusing on the code sequence, this code only catches errors and returns an error code to the caller function. In your conduit, you must handle the specific error codes returned by each Sync Manager function. The connection with the handheld may be

interrupted at any time and, so you must check for errors after each Sync Manager function call. Read about the Sync Manager error codes in the *C/C++ Sync Suite Reference* for more information.